Xu Han

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Photonics M.Sc. student and LMU astrophysics graduate focused on astronomical instrumentation, wavefront sensing and optical-system performance, combining optical modelling, experimental apparatus development and astronomical data analysis.

Education

M.Sc. Photonics | Munich University of Applied Sciences *Mar 2026 - Present*
Selected coursework: Physical Optics and Modelling (**Zemax optical design**); Micro and Fiber Optics; Optical Measurement Technology; Semiconductor Optics and Components.

M.Sc. Astrophysics | LMU Munich *Apr 2022 - Aug 2024*
Grade: 1.55 (German scale). **Selected coursework:** Essentials of Advanced Astrophysics; Cosmology and Structure Formation; Statistical Methods; Instrumental Laboratory Courses.
Degree certificate

B.Sc. Physics | University of Duisburg-Essen *Oct 2018 - Mar 2022*
Grade: 1.8 (German scale).
Degree certificate

B.Eng. Electrical Engineering and Automation *Sep 2013 - Jul 2017*
Hubei University of Technology
Electronics, control systems and embedded systems.

Selected Research and Engineering Projects

Shack-Hartmann Wavefront Sensing & Adaptive Optics Simulation

Independent project (ongoing development) *Mar 2026 - Present*
Interactive explorer | Source code

- Developed a modular Python SCAO framework connecting atmospheric turbulence, detector-level Shack-Hartmann sensing, calibrated reconstruction, deformable-mirror correction and delayed feedback.
- Investigated detector-noise and gain/latency trade-offs using residual wavefront error, propagated-PSF Strehl and encircled energy.
- Diagnosed a reconstruction failure associated with measurement dropout in a 2.2 m NGS model; stronger TSVD regularisation kept reconstruction usable throughout the matched two-second run.
- Cross-checked selected optical components against HCIPy and analytic references; documented reproducible experiments, model limitations and an interactive explorer of recorded results.

Weak Lensing Void Finder *Mar 2023 - Aug 2024*

Master's thesis | LMU Munich
Supervisors: Prof. Joseph Mohr and Dr. Chris Davies.

- Developed and parameter-tested a watershed finder for cosmic voids in 2D full-sky weak-lensing convergence maps.
- Applied the analysis to DES Y3 weak-lensing maps; characterised void abundance, effective radii, ellipticities and convergence profiles.

Experimental Collisional Charging Apparatus *Mar 2021 - Aug 2021*

Bachelor's thesis | University of Duisburg-Essen
Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.

- Designed the mechanical apparatus in CAD, fabricated components by 3D printing and assembled the setup; integrated and debugged electronic charge-measurement modules.
- Conducted measurements and analysed grain-size dependence of collisional charging, testing a dust-collision growth hypothesis relevant to early planet formation.

Laboratory and Instrumentation Training

- **Astronomical instrumentation laboratory:** Tested static and dynamic correction on LMU's ALPAO bench in July 2022 using a Shack-Hartmann sensor and deformable mirror. Other labs covered CCD imaging and reduction, optical interferometry and radio measurements of Galactic rotation.
- **Photonics laboratory:** Measured laser beam quality (M^2), shaped ultrashort pulses and characterised them by autocorrelation. Fibre experiments covered OTDR, polarisation, Bragg gratings and coupling. Also worked on APD measurements, laser spectra, laser interferometry and nonlinear optics.
- **Observatory experience:** Took and processed images with my supervisor at Wendelstein (43 cm).

Additional Independent Projects

Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fiber coupling, spectral SNR and radial-velocity precision.

OWL Optical Reconstruction

Personal project extending a hands-on Zemax lab in Physical Optics and Modeling

- Reproduced the 5 arcmin RMS wavefront error to 0.3% and traced a 27 percentage-point ensquared-energy discrepancy to the choice of focal surface.

Work Experience and Career Break

Python Developer | BaryonBib *Dec 2024 - Feb 2025*
MAX!mize start-up team, MPA, Garching

- Implemented and tested LLM features for the team's centralised Python bibliography platform.

Medical leave *Jan 2025 - Dec 2025*
Recovery and rehabilitation following an accident and surgery.

Talks, Workshops and Schools

MAVIS 2026 workshop | Participant *Sep 2026*
Tonale Winter School on Cosmology | Participant *Dec 2024*
ORIGINS Lensing Day *Nov 2024*
Contributed talk: Watershed on the curved sky
Cambridge-LMU Cosmology Meeting | Participant *Dec 2023*
Advanced Astrophysical Seminar | LMU Munich *Nov 2022*
Exploring Telescope Optics: How Do the Largest Optical Systems Work? 40-minute course presentation supervised by Dr. Frank Grupp.

Technical Skills

Optical modelling: Wavefront sensing, Fourier PSFs, calibration, regularised reconstruction and closed-loop diagnostics; HCIPy and TIPTOP.
Zemax OpticStudio: telescope prescription reconstruction, PSF and encircled-energy analysis
Scientific computing: Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn); MATLAB; Mathematica; statistical analysis and MCMC.
Experimental methods: CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment and interferometry.
Research tools: Linux, Git, Jupyter and LaTeX.

Languages and Interests

Languages: Chinese (native); English (fluent); German (fluent); French (beginner).
Interests: Astronomical photography and alpine mountaineering.

Academic Reference

Prof. Joseph Mohr, M.Sc. thesis supervisor